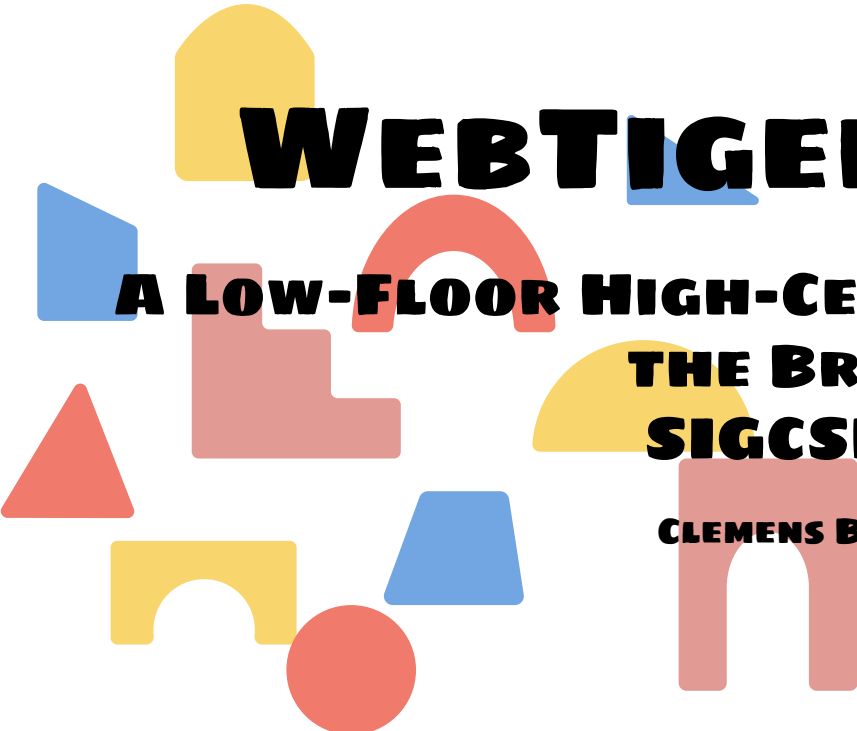


A collection of abstract geometric shapes in yellow, blue, red, and pink, scattered around the text. The shapes include a yellow semi-circle, a blue trapezoid, a red arch, a pink L-shape, a yellow semi-circle, a red triangle, a pink L-shape, a blue trapezoid, a yellow arch, a red circle, and a pink arch.

WEBTIGERPYTHON

**A LOW-FLOOR HIGH-CEILING PYTHON IDE FOR
THE BROWSER
SIGCSE 2026**

CLEMENS BACHMANN

WHO AM I

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- University: ETH Zürich (Switzerland)





AUTHORS OF THIS PAPER

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EDUCATIONAL IDE - REQUIREMENTS

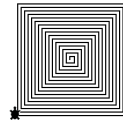
- Simple
- Fast
- Works on any device
- Robotics
- Visual Output

High Ceiling

matplotlib

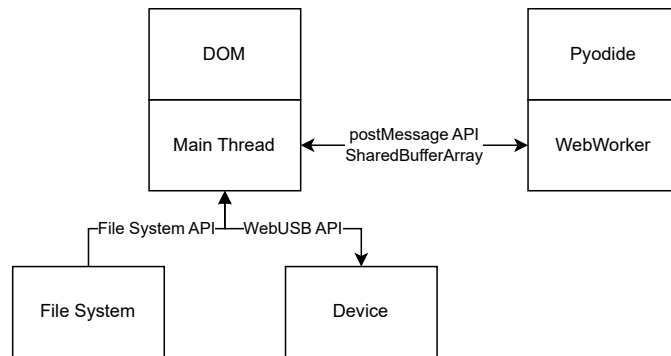


Low Floor



IMPLEMENTATION WEBTIGERPYTHON

- Works on any device -> Web Based
- Web Worker to execute code
- Web APIs for robotics / files



LOW FLOOR 1 - TURTLE

- Reimplementation from scratch
 - Tkinter does not work in the browser
- PixiJS

 WebTigerPython
Tiger.Jython for the Web

```
1 from gturtle import *
2 makeTurtle()
3
4 for color in ["red",
5              "blue",
6              "darkgreen",
7              "orange"]:
8     setPenColor(color)
9     repeat 4:
10        forward(100)
11        right(90)
12        right(90)
13
```

ⓘ Loading Python

Supported by

  Algorithms and Didactics

→

LOW FLOOR 2 - PHYSICAL COMPUTING

- WebUSB
 - Only in Chrome
- Simulation

 WebTigerPython
TigerJython for the Web

```
1 from mbrobot_plusV3 import *
2
3 forward()
4 v repeat:
5     d = getDistance()
6     print(d)
7 v if d < 20:
8     stop()
9     delay(200)
10
```

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HIGH CEILING 1 - SCIENTIFIC PYTHON

- Numpy / Scipy / Matplotlib
 - Thanks to Pyodide
- Use agg (Anti-Grain Geometry) to render
- Limitation: no direct access to the DOM
 - No events
 - No animations

 WebTigerPython
Tiger.Jython for the Web

1 `import numpy as np`

2 `import matplotlib.pyplot as plt`

3

4 `x = np.arange(0, 10.01, 0.01)`

5 `f1 = np.sin(x)`

6 `f2 = np.cos(x)`

7 `f3 = 0.01 * x**2 + 0.15 * x - 1`

8

9 `plt.plot(x, f1, color="red")`

10 `plt.plot(x, f2, color="blue")`

11 `plt.plot(x, f3, color="green")`

12

13 `plt.show()`

14

Loading Python

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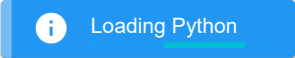
 

→

HIGH CEILING 1 - INTERACTIVE PROGRAMMING

- Input
- Inputlisteners
 - Keyboard
 - Mouse

 WebTigerPython
TigerJython for the Web



```
1 from turtle import *
2 from time import *
3 from random import randrange
4
5
6 breite = Screen().window_width()
7 hoehe = Screen().window_height()
8 fps = 10
9 farbeSchlange = "green"
10 farbeEssen = "red"
11
12 spc("white")
13 dot(1000)
14
15 def createFood():
16     pos = getPos()
17     done = False
18     spc(farbeEssen)
19     while not done:
20         x = randrange(-breite//20*10
21         y = randrange(-hoehe//20*10,
22         setPos(x,y)
23         color = getPixelColorStr()
24         if color == "white":
25             dot(8)
26             setPos(pos)
27             spc(farbeSchlange)
28             done = True
29     }
30 def richtungsWechsel(x):
31     if x == "a":
```

Supported by





HIGH CEILING

2 - SCIENTIFIC

PYTHON

- Numpy / Scipy / Matplotlib
- Use agg to render within python
- Limitation
 - No events
 - No animations

 WebTigerPython
TigerJython for the Web

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 x = np.arange(0, 10.01, 0.01)
5 f1 = np.sin(x)
6 f2 = np.cos(x)
7 f3 = 0.01 * x**2 + 0.15 * x - 1
8
9 plt.plot(x, f1, color="red")
10 plt.plot(x, f2, color="blue")
11 plt.plot(x, f3, color="green")
12
13 plt.show()
14
```

Loading Python

Supported by

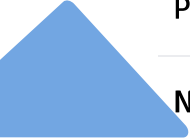
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EVALUATION 1 - COMPUING BENCHMARK

MANDELBROT SET COMPUTATION

Test Case: 200×200 pixel image with 4,000,000 iterations of $z' = z^2 + c$

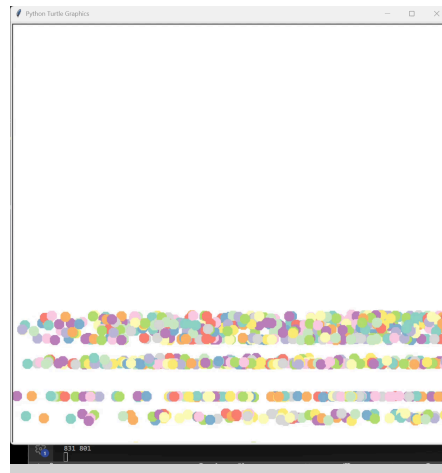
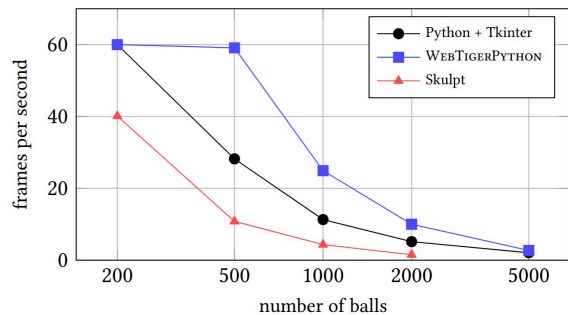


Platform	Execution Time	Performance vs CPython
Native Python	0.16 s	Baseline
WebTigerPython	0.38 s	~2.4× slower
PyodideU	118 s	~738× slower
trinket.io (Skulpt)	43 s	~269× slower

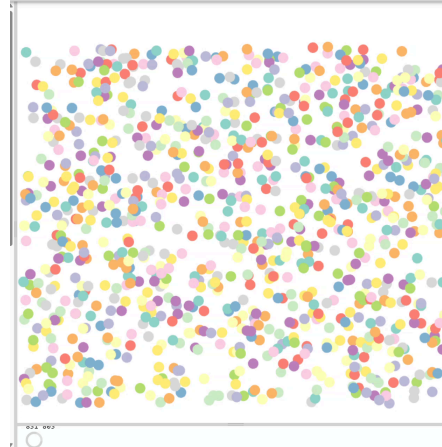


EVALUATION 2 - GRAPHICS BENCHMARK

- WTP outperforms Native Python in this benchmark



Native Python



WebTigerPython



EVALUATION 3 - TEACHER SURVEY

- Userbase 12 years and above
 - Most Users 15 - 17 years
 - URL Sharing was liked
 - Easy integratable into other tools
 - Simple
- 



MORE TO EXPLORE!

- 
- Debugger
 - Robotics Simulator
 - Pygame integration
 - Music
- 



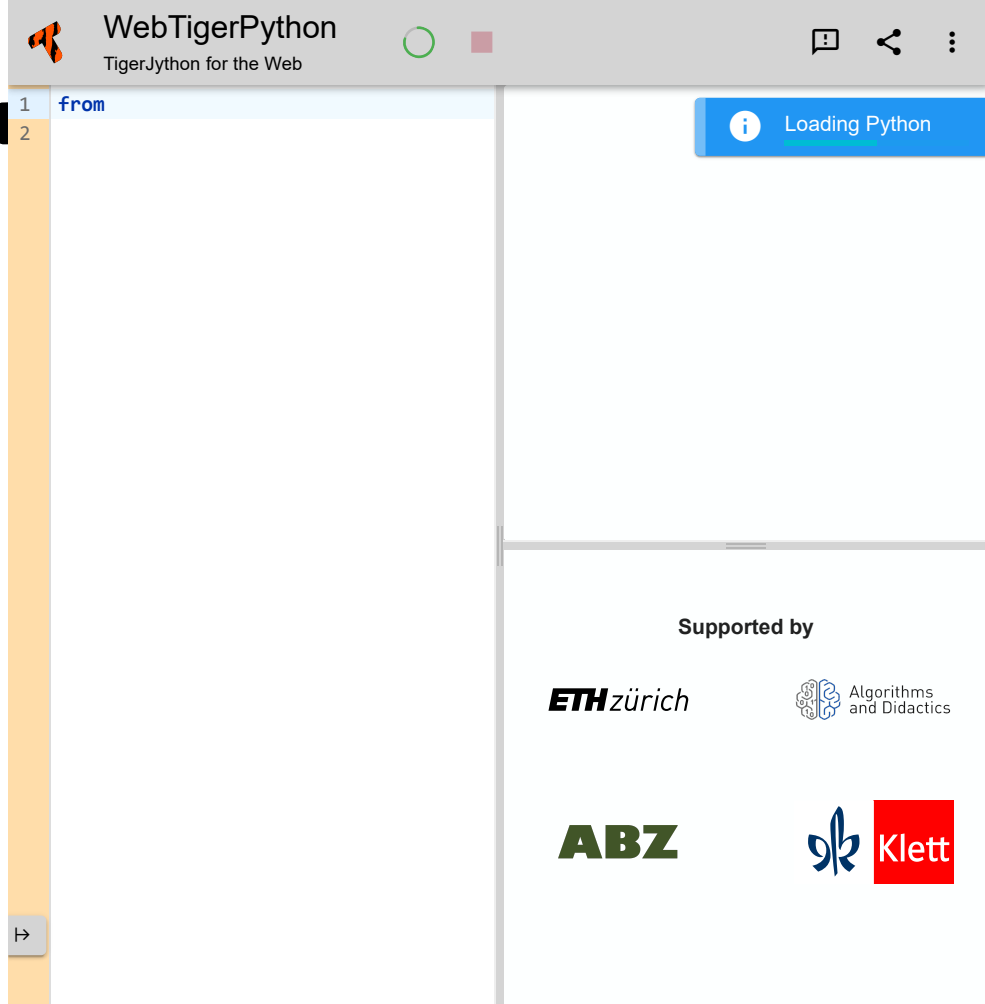
FUTURE WORK

- Multifile
- Robots over WiFi ...



WEBTIGERPYTHON

- Educational IDE
- Low Floor
 - Turtle
 - Robotics
- High Ceiling
 - Interactive Programming
 - Scientific Programming
- Benchmarks
 - 2-3 Times slower than Native Python
 - WTP turtle is faster in our Benchmark



The screenshot shows the WebTigerPython web IDE interface. At the top, the browser tab is titled "WebTigerPython" with the subtitle "TigerJython for the Web". The interface is divided into three main sections: a code editor on the left, a console/output area in the middle, and a sidebar on the right. The code editor shows a Python file with line numbers 1 and 2, and the text "from". The console area is currently empty. The sidebar on the right contains a blue button with an information icon and the text "Loading Python". Below this, there is a section titled "Supported by" which lists several logos: ETH zürich, Algorithms and Didactics, ABZ, and Klett.

WebTigerPython
TigerJython for the Web

1 from
2

Loading Python

Supported by

ETH zürich Algorithms and Didactics

ABZ Klett